WELCOME

ANJU PRAKASH P

DADS JULY OFFLINE BATCH

MILESTONE PROJECT - 1

LAPTOP

- COMPANY
- RAM
- MEMORY
- TYPENAME
- PRICE
- INCHES
- OPSYS
- WEIGHT
- SCREENRESOLUTION
- GPU
- CPU

INTRODUCTION

- Analyze laptop characteristics
- Evaluate laptop trend
- Examine laptop pricing

OVERVIEW

READING DATA → UNDERSTANDING → CLEANING → PROCESSING

OUTLIER → DERIVED MATRIX → PLOTS → EDA

Reading Data

Out[3]:

	Unnamed: 0	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	0.0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	1.0	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	2.0	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000
3	3.0	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195.3360
4	4.0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8080
...	...	...	...	...	...	...	...	...	...	...	...	...
1298	1298.0	Lenovo	2 in 1 Convertible	14	IPS Panel Full HD / Touchscreen 1920x1080	Intel Core i7 6500U 2.5GHz	4GB	128GB SSD	Intel HD Graphics 520	Windows 10	1.8kg	33992.6400
1299	1299.0	Lenovo	2 in 1 Convertible	13.3	IPS Panel Quad HD+ / Touchscreen 3200x1800	Intel Core i7 6500U 2.5GHz	16GB	512GB SSD	Intel HD Graphics 520	Windows 10	1.3kg	79866.7200
1300	1300.0	Lenovo	Notebook	65.4	1366x768	Intel Celeron Dual Core N3050 1.6GHz	2GB	64GB Flash Storage	Intel HD Graphics	Windows 10	1.5kg	12201.1200
1301	1301.0	HP	Notebook	15.6	1366x768	Intel Core i7 6500U 2.5GHz	6GB	1TB HDD	AMD Radeon R5 M330	Windows 10	2.19kg	40705.9200
1302	1302.0	Asus	Notebook	15.6	1366x768	Intel Celeron Dual Core N3050 1.6GHz	4GB	500GB HDD	Intel HD Graphics	Windows 10	2.2kg	19660.3200

1303 rows × 12 columns

UNDERSTANDING DATA

```
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1273 non-null   float64
 1   Company           1273 non-null   object
 2   TypeName          1253 non-null   object
 3   Inches            1220 non-null   object
 4   ScreenResolution  1273 non-null   object
 5   Cpu               1273 non-null   object
 6   Ram               1259 non-null   object
 7   Memory            1273 non-null   object
 8   Gpu               1273 non-null   object
 9   OpSys             1273 non-null   object
 10  Weight            1259 non-null   object
 11  Price             1273 non-null   float64
dtypes: float64(2), object(10)
```

```
Unnamed: 0          30
Company             30
TypeName            50
Inches              83
ScreenResolution    30
Cpu                 30
Ram                 44
Memory              30
Gpu                 30
OpSys               30
Weight              44
Price               30
```

Initial Exploration

```
#droping unwanted column
df.drop(columns=['Unnamed: 0'],inplace=True)
```

```
#removing empty rows
df.dropna(axis='rows',how='all',inplace=True)
```

PERCENTAGE OF NULL VALUES

```
Inches              6.37
TypeName            3.84
Ram                 3.38
Weight              3.38
Company             2.30
ScreenResolution    2.30
Cpu                 2.30
Memory              2.30
Gpu                 2.30
OpSys               2.30
Price               2.30
dtype: float64
```

Initial Exploration

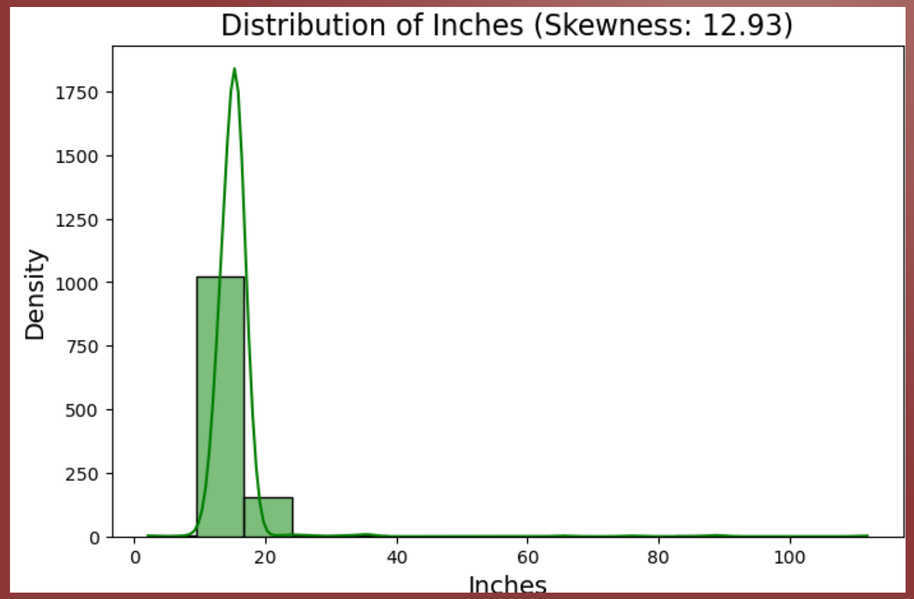
Unnamed: 0	30
Company	30
TypeName	50
Inches	83
ScreenResolution	30
Cpu	30
Ram	44
Memory	30
Gpu	30
OpSys	30
Weight	44
Price	30

→

Company	0
TypeName	20
Inches	53
ScreenResolution	0
Cpu	0
Ram	14
Memory	0
Gpu	0
OpSys	0
Weight	14
Price	0

DATA CLEANING

Company	0
TypeName	20
Inches	53
ScreenResolution	0
Cpu	0
Ram	14
Memory	0
Gpu	0
OpSys	0
Weight	14
Price	0

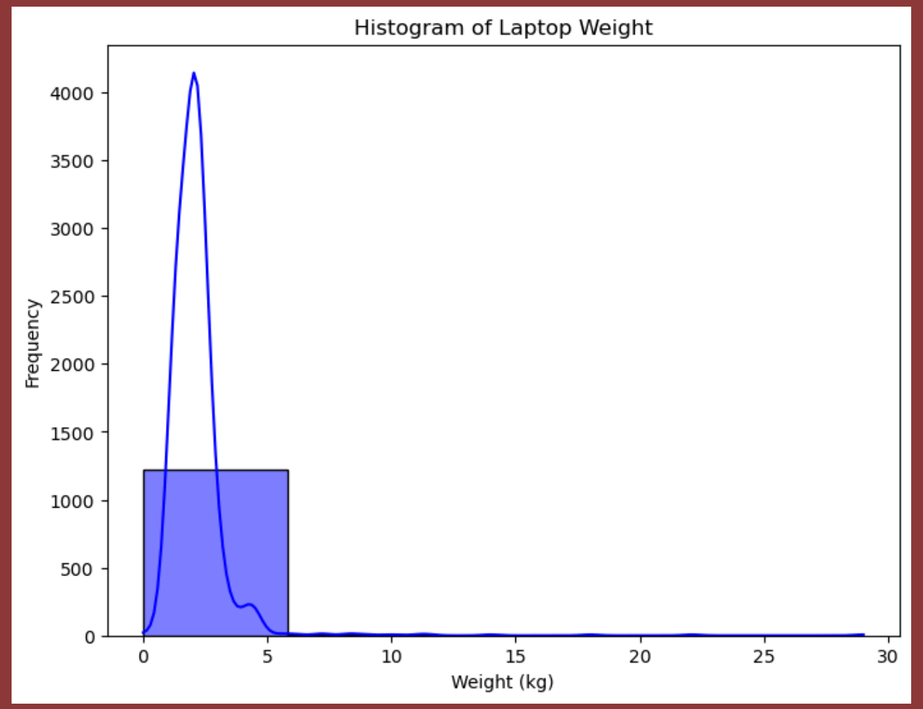


INCHES

- Dtype:Convert object to float
- The distribution of the 'Inches' : right-skewed.
- Removed ‘?’ by droping the column
- Fill with :median

WEIGHT

- Dtype:Convert object to float
- The distribution of the 'Weight' : right-skewed.
- Removed ‘?’ by droping the column and removed unit ‘kg’ for the conversion
- Fill with :median



TYPE NAME

Company	no of columns	Typename filled
Acer	1	Notebook
Asus	1	Notebook
Dell	6	Notebook
Fujitsu	1	Notebook
HP	4	Notebook
Lenovo	5	Notebook
Tosiba	2	Notebook

RAM

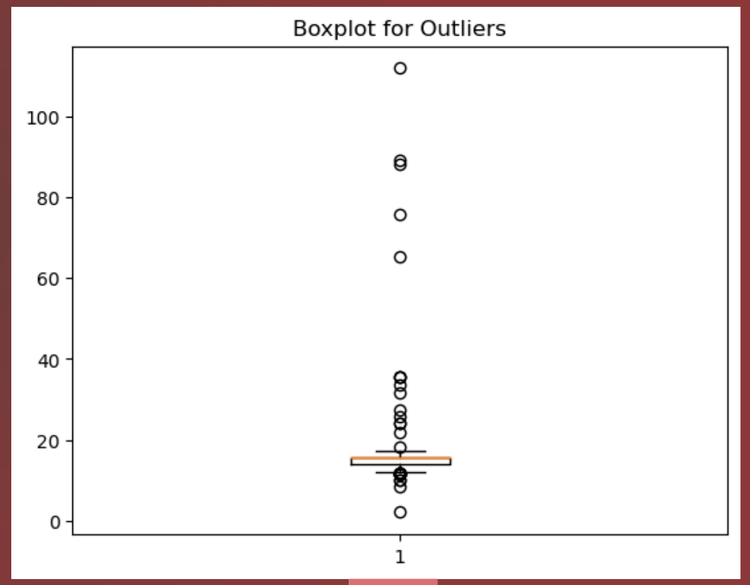
Typename	no of columns	RAM
2 in 1 convertible	1	8
Gaming	1	16
Notebook	9	4
Ultrabook	3	8

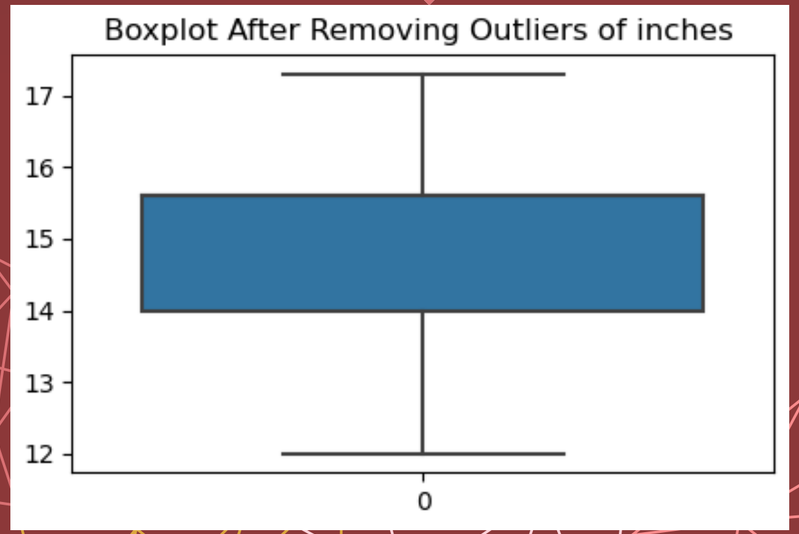
CLEANED DATA

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83	135195.3360
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37	96095.8080
...	...	...	...	...	...	...	...	...	...	...	...
1270	Lenovo	2 in 1 Convertible	14.0	IPS Panel Full HD / Touchscreen 1920x1080	Intel Core i7 6500U 2.5GHz	4GB	128GB SSD	Intel HD Graphics 520	Windows 10	1.80	33992.6400
1271	Lenovo	2 in 1 Convertible	13.3	IPS Panel Quad HD+ / Touchscreen 3200x1800	Intel Core i7 6500U 2.5GHz	16GB	512GB SSD	Intel HD Graphics 520	Windows 10	1.30	79866.7200
1272	Lenovo	Notebook	14.0	1366x768	Intel Celeron Dual Core N3050 1.6GHz	2GB	64GB Flash Storage	Intel HD Graphics	Windows 10	1.50	12201.1200
1273	HP	Notebook	15.6	1366x768	Intel Core i7 6500U 2.5GHz	6GB	1TB HDD	AMD Radeon R5 M330	Windows 10	2.19	40705.9200
1300	Lenovo	Notebook	65.4	1366x768	Intel Celeron Dual Core N3050 1.6GHz	2GB	64GB Flash Storage	Intel HD Graphics	Windows 10	1.50	12201.1200

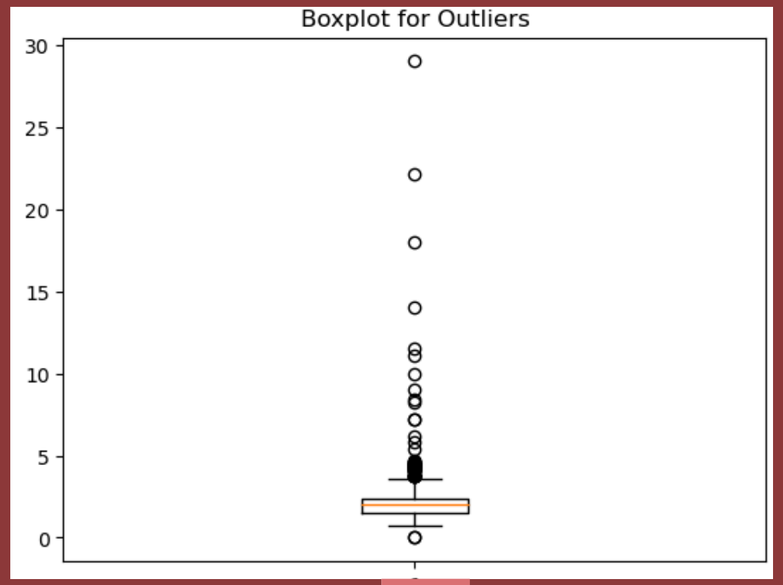
1244 rows × 11 columns

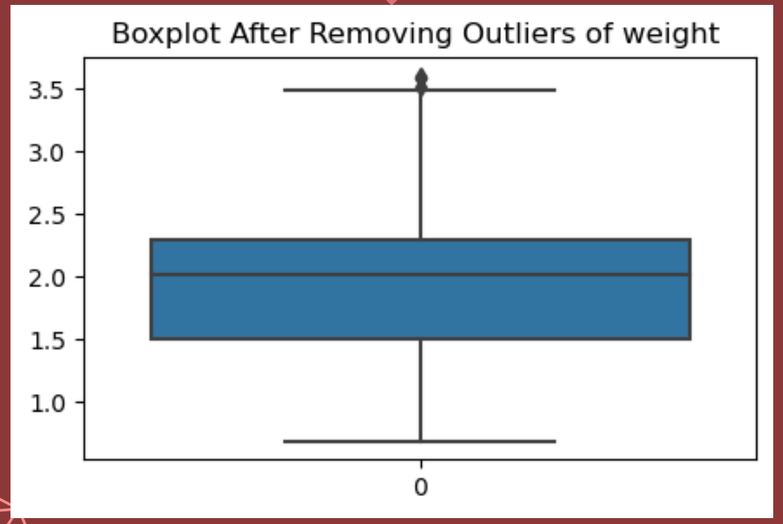
INCHES



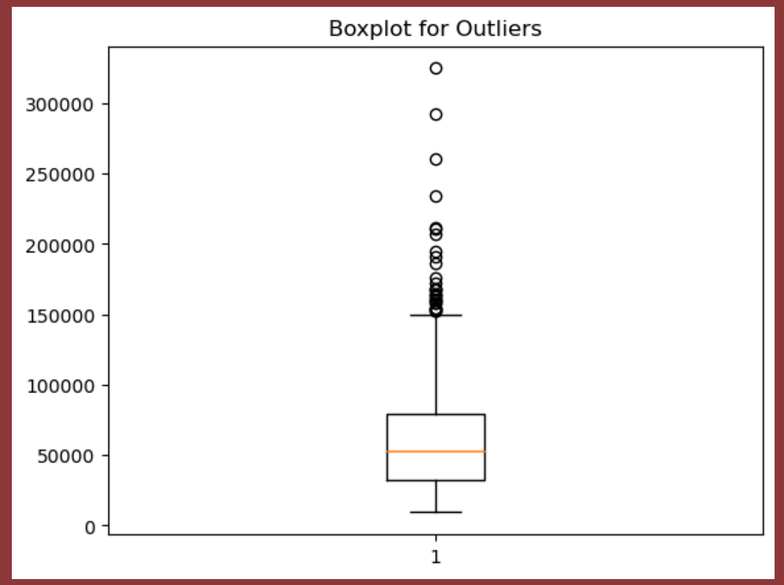


WEIGHT





PRICE



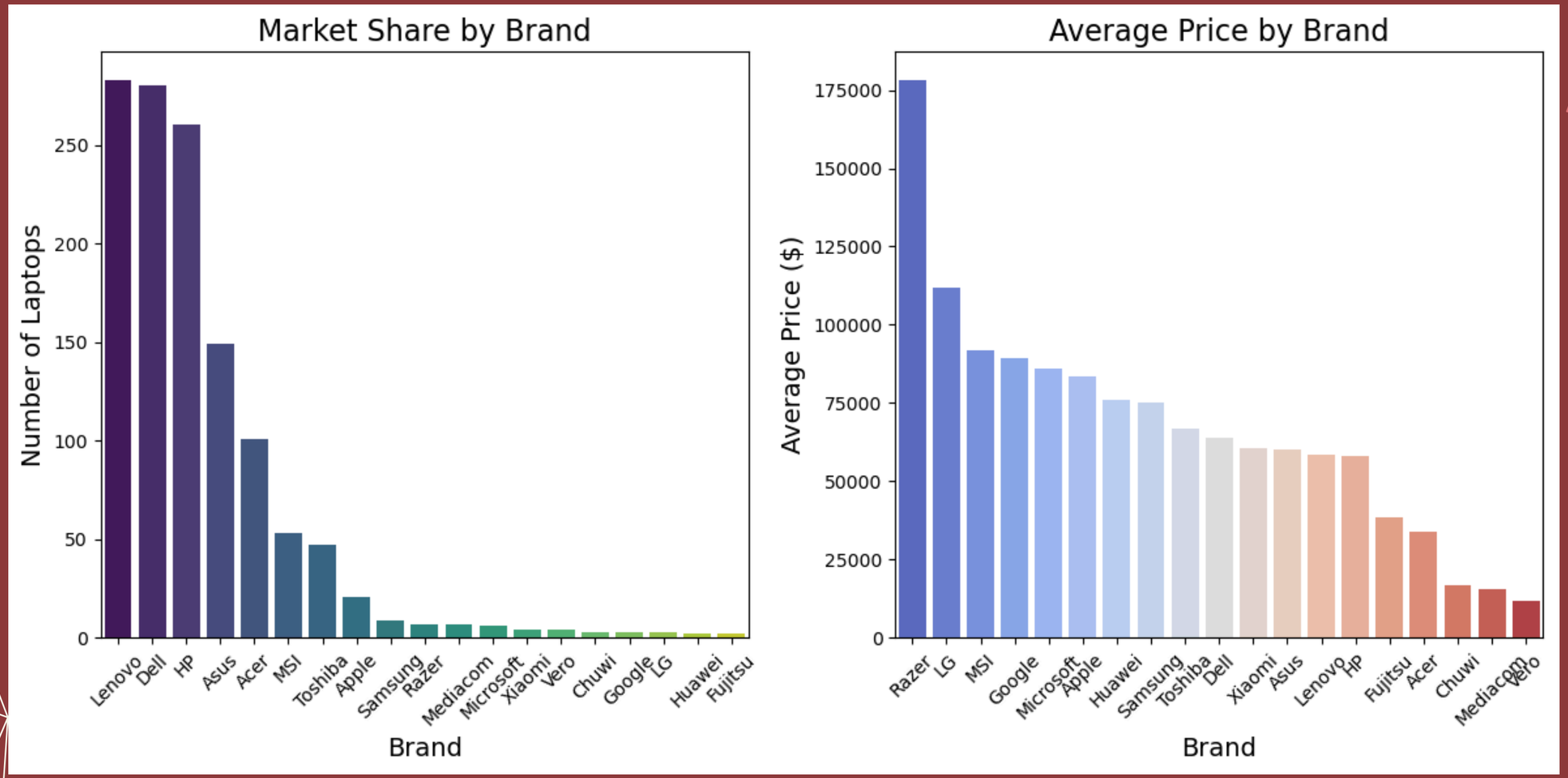
?

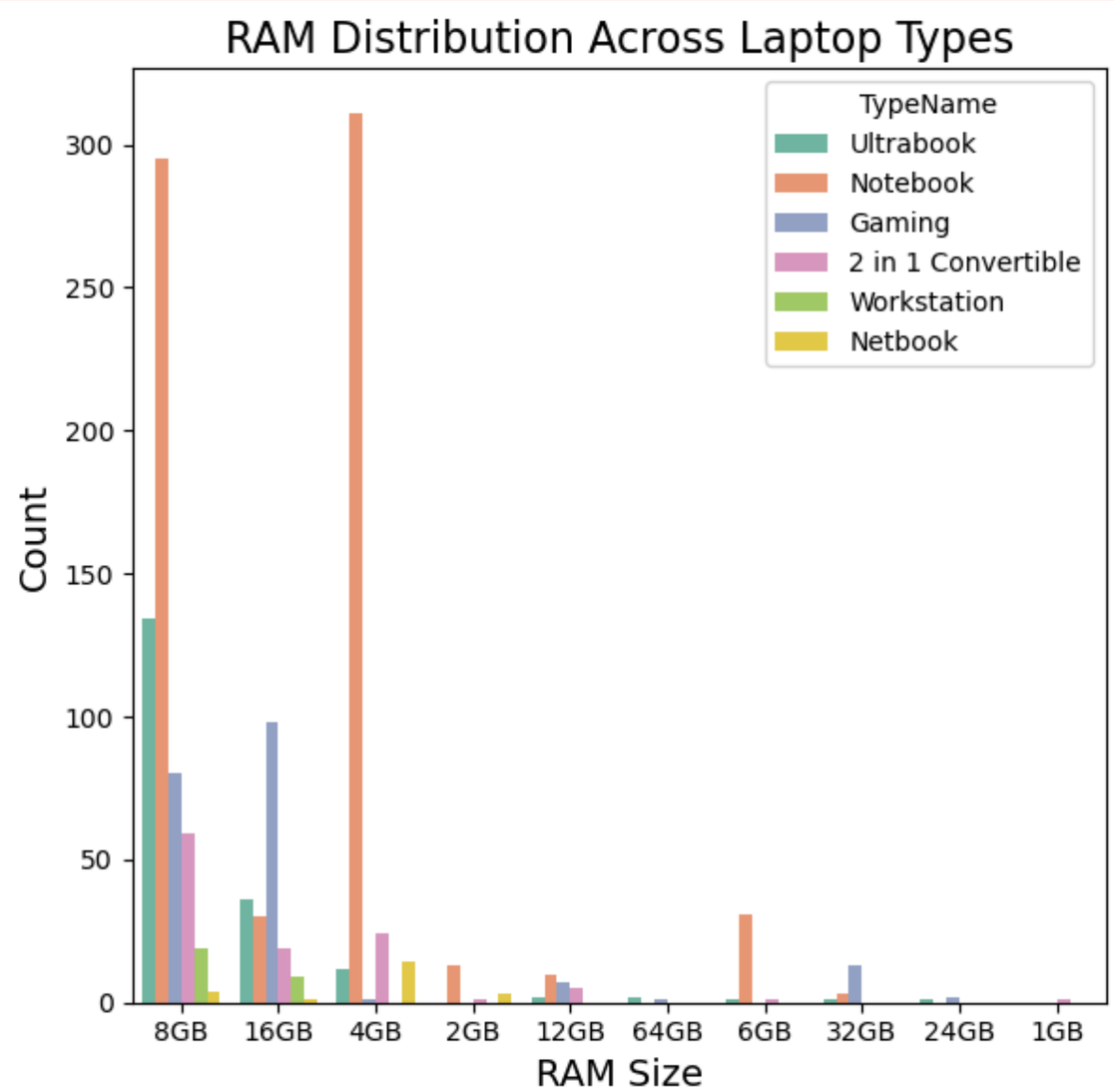
DERIVED MATRIX

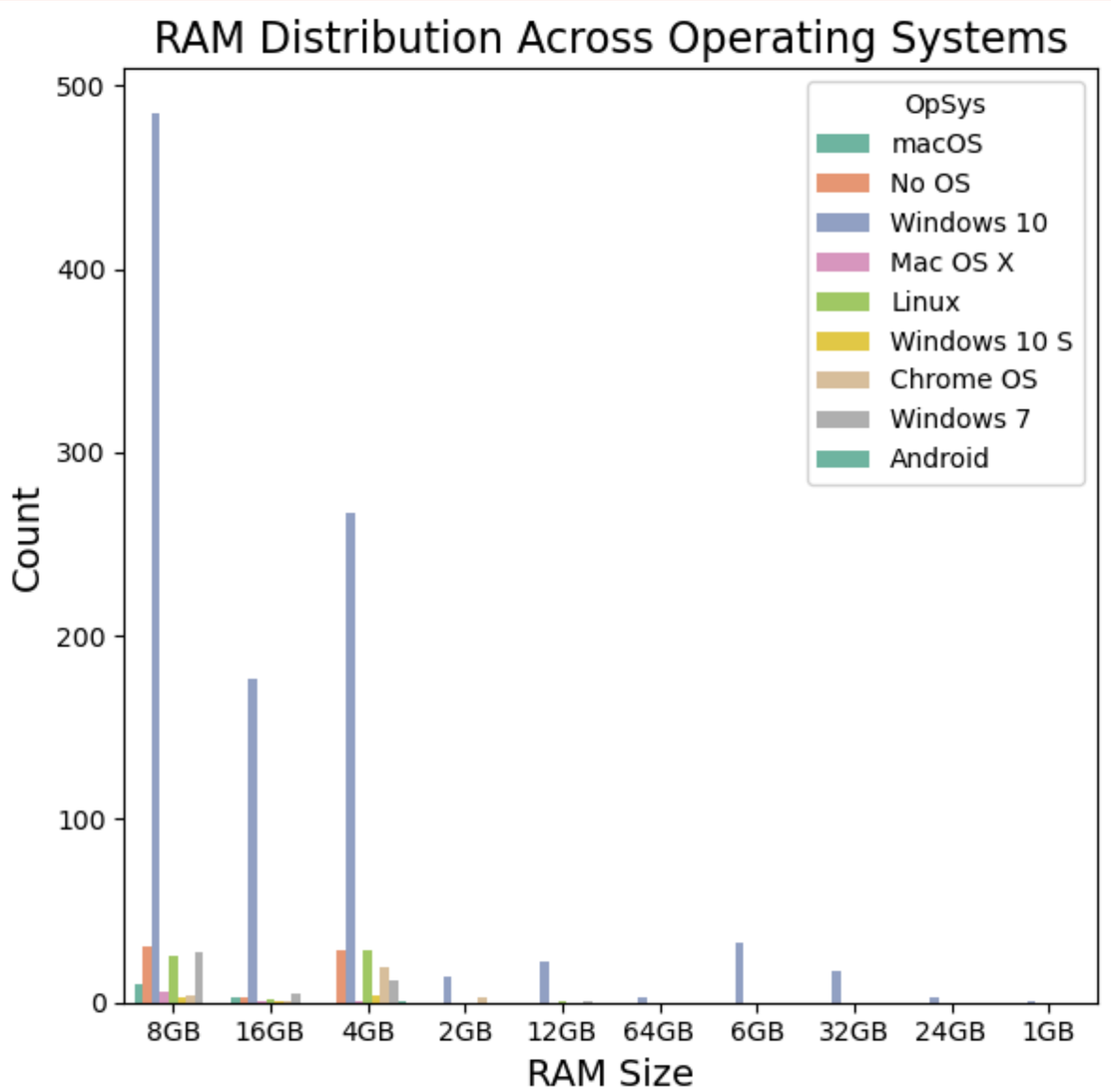
	TypeName	OpSys	MeanPrice	MedianPrice	StdPrice	Count
0	2 in 1 Convertible	Android	29250.720000	29250.7200	NaN	1
1	2 in 1 Convertible	Chrome OS	33223.885714	32767.2000	14120.803575	7
2	2 in 1 Convertible	Windows 10	70676.305093	69210.7200	31003.590282	101
3	2 in 1 Convertible	Windows 7	109244.246400	109244.2464	NaN	1
4	Gaming	Linux	45088.200000	46007.2800	6041.330995	4
5	Gaming	No OS	54327.840000	53173.4400	10464.163515	9
6	Gaming	Windows 10	94605.732724	80133.1200	43394.055609	189
7	Netbook	Chrome OS	15503.814000	15528.4560	3360.238926	8
8	Netbook	Windows 10	48968.760000	34552.0800	34420.195542	12
9	Netbook	Windows 10 S	18115.200000	18115.2000	NaN	1
10	Netbook	Windows 7	85194.720000	85194.7200	NaN	1
11	Notebook	Chrome OS	19067.728000	18594.7200	6690.790613	9
12	Notebook	Linux	31253.724141	27864.9072	12712.277946	51
13	Notebook	No OS	27679.804438	26101.8720	11489.611434	53
14	Notebook	Windows 10	43841.205036	39373.9200	24426.023570	557
15	Notebook	Windows 10 S	14811.307200	14811.3072	NaN	1
16	Notebook	Windows 7	71848.273745	63669.6000	31557.359453	22
17	Ultrabook	Chrome OS	89386.080000	83063.5200	25216.998664	3
18	Ultrabook	Linux	58554.720000	58554.7200	0.000000	2
19	Ultrabook	Mac OS X	67285.780200	62017.9200	19933.945968	8
20	Ultrabook	No OS	53274.672000	53274.6720	NaN	1
21	Ultrabook	Windows 10	82997.117471	83063.5200	25548.152565	143
22	Ultrabook	Windows 10 S	85903.788000	83622.9600	31884.406736	6
23	Ultrabook	Windows 7	91422.127385	91288.3536	23272.951728	13
24	Ultrabook	macOS	93220.327385	80908.3440	31352.263798	13
25	Workstation	Windows 10	115740.170640	108102.4560	27035.893344	20
26	Workstation	Windows 7	135040.824000	109969.9200	58922.496775	8

EXPLORATORY DATA ANALYSIS THROUGH VISUALISATION

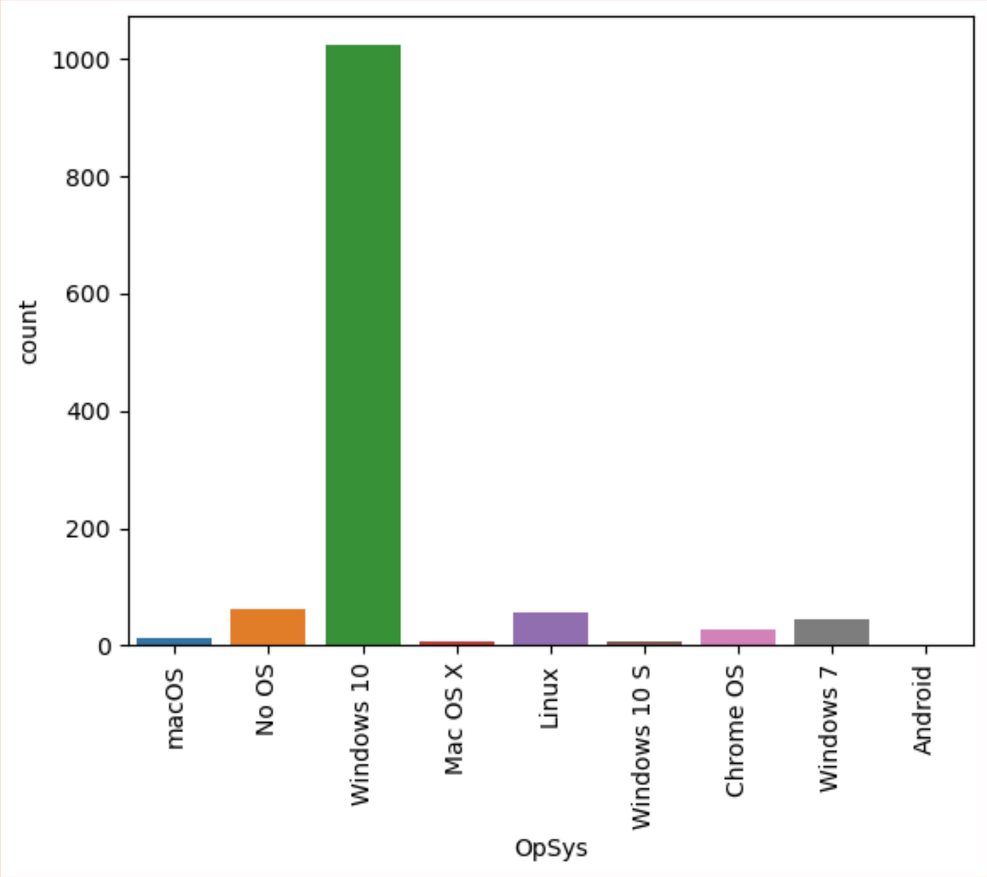
Identify which brands dominate the market and have premium pricing.

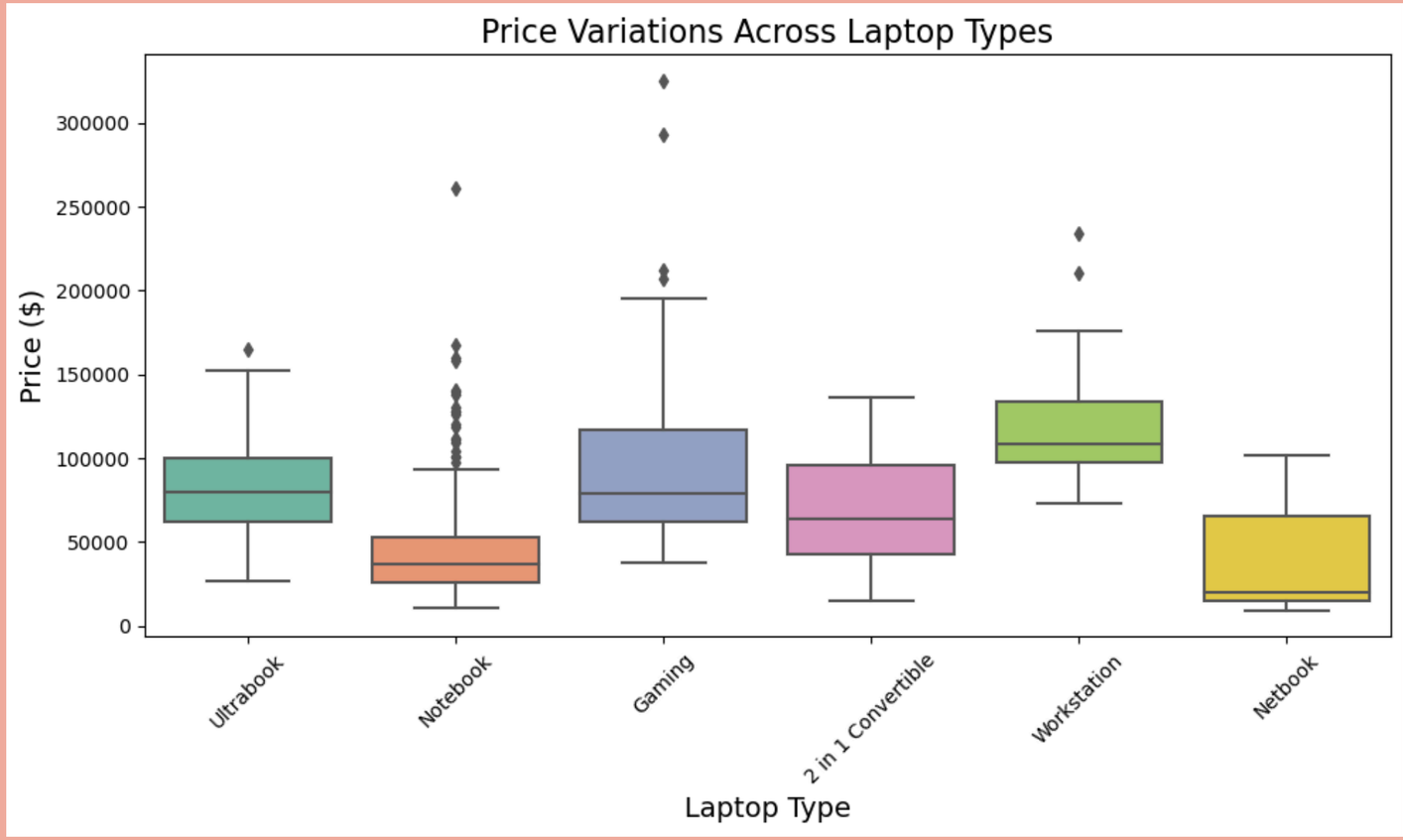




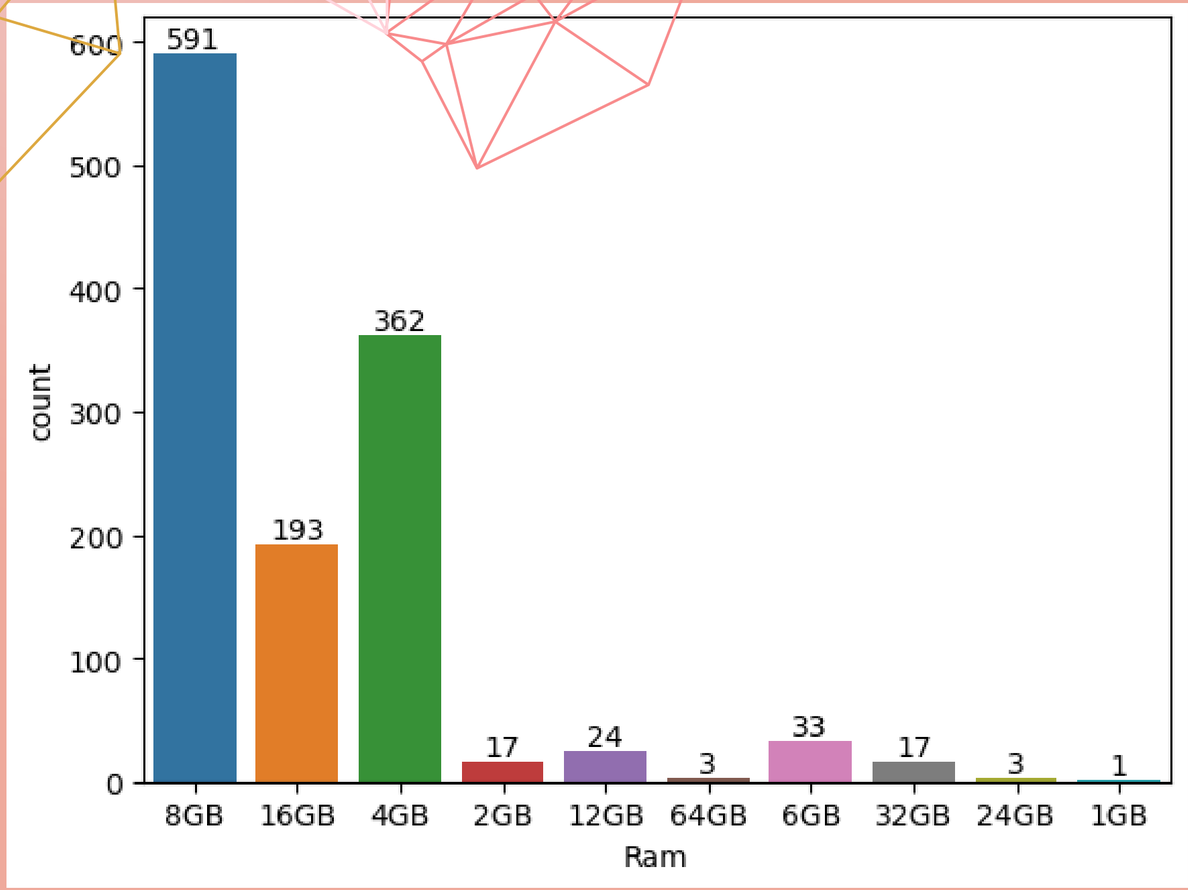


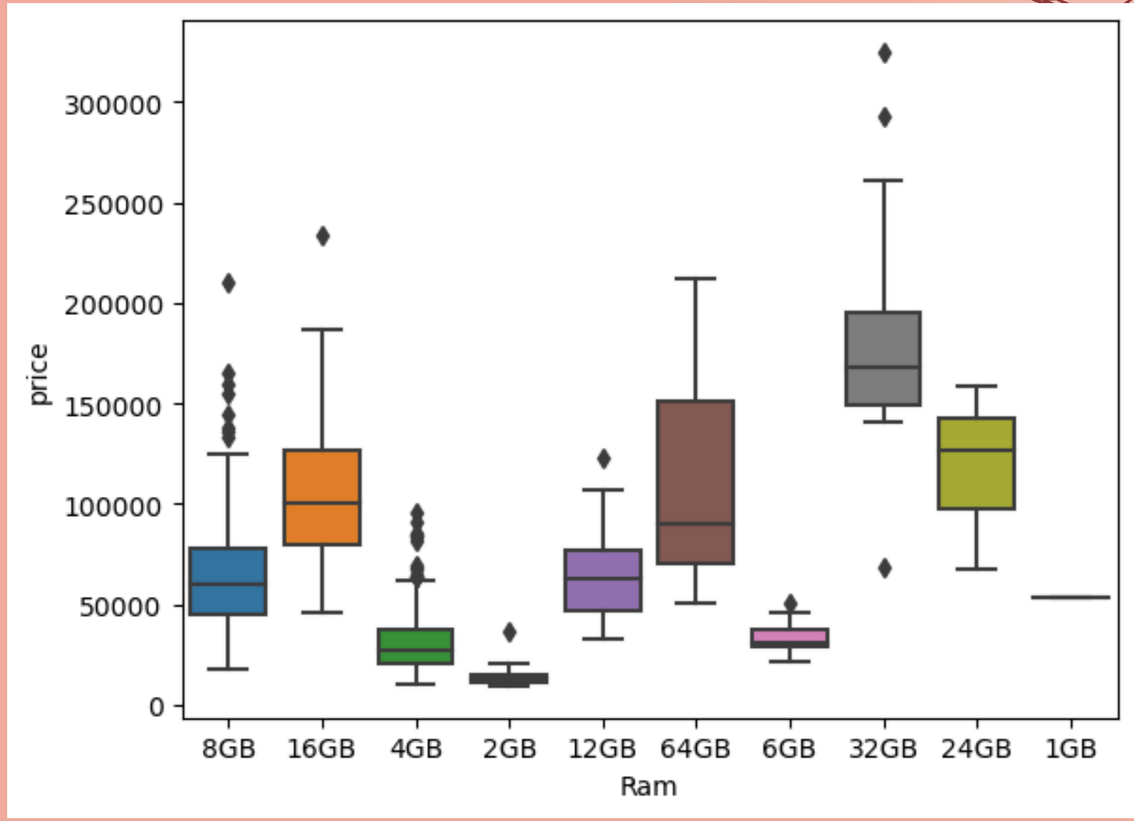
Check for trends between RAM size and laptop types or operating systems



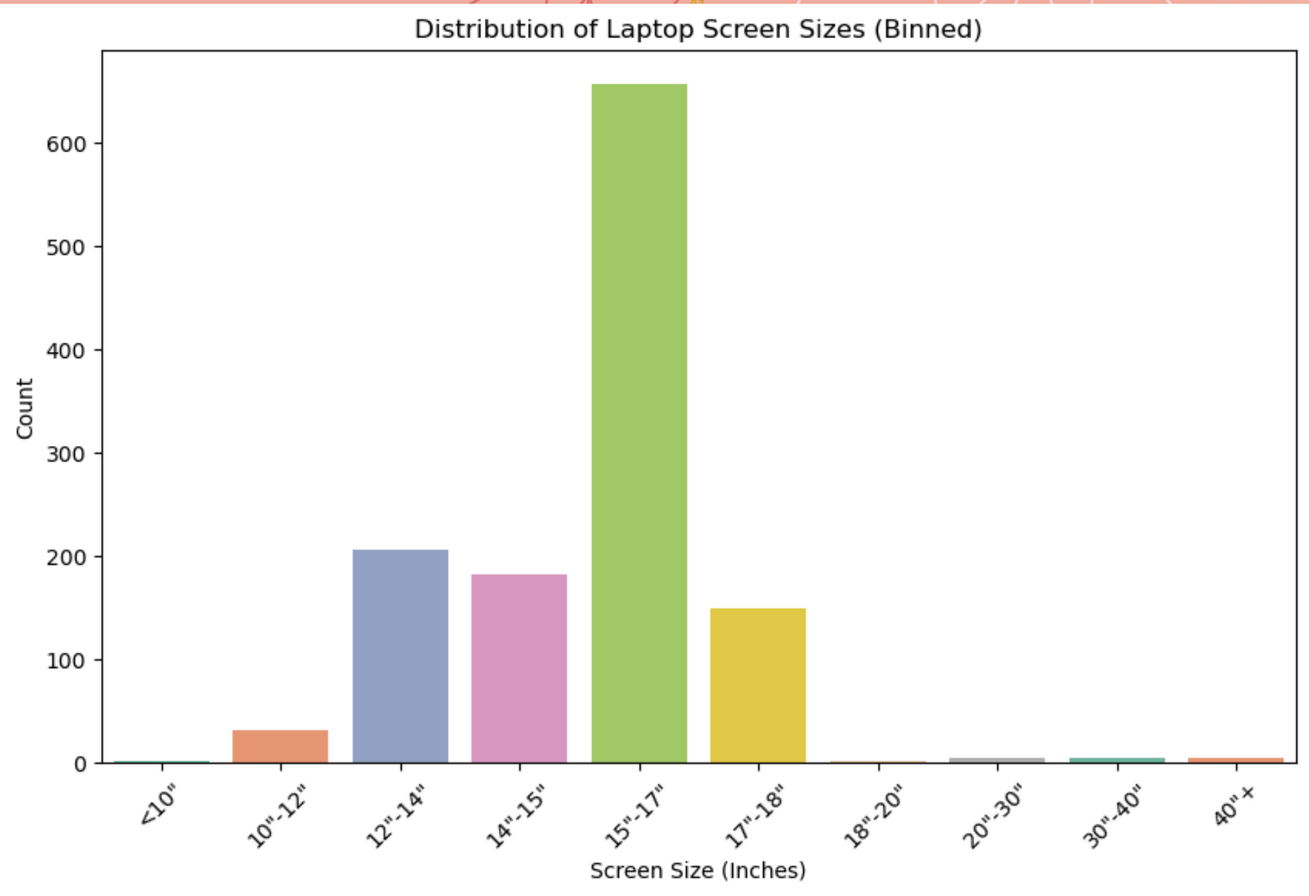


Idendifing the most used laptop type

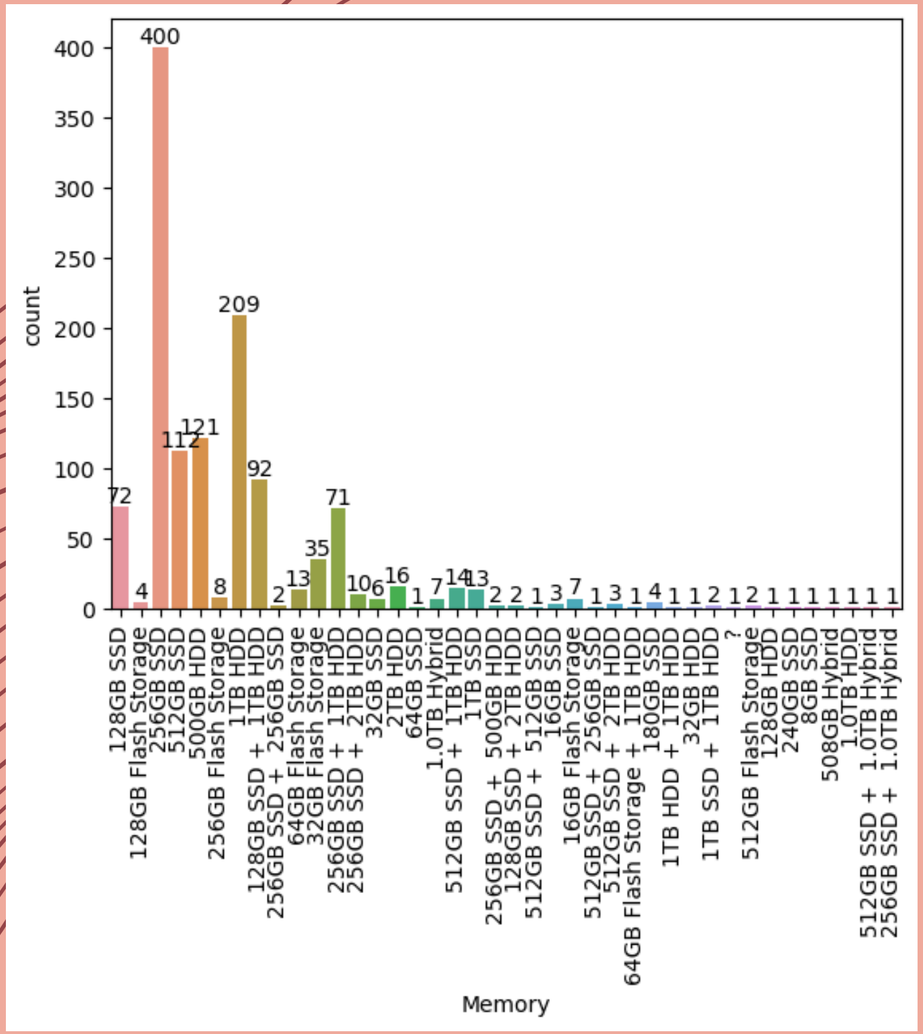


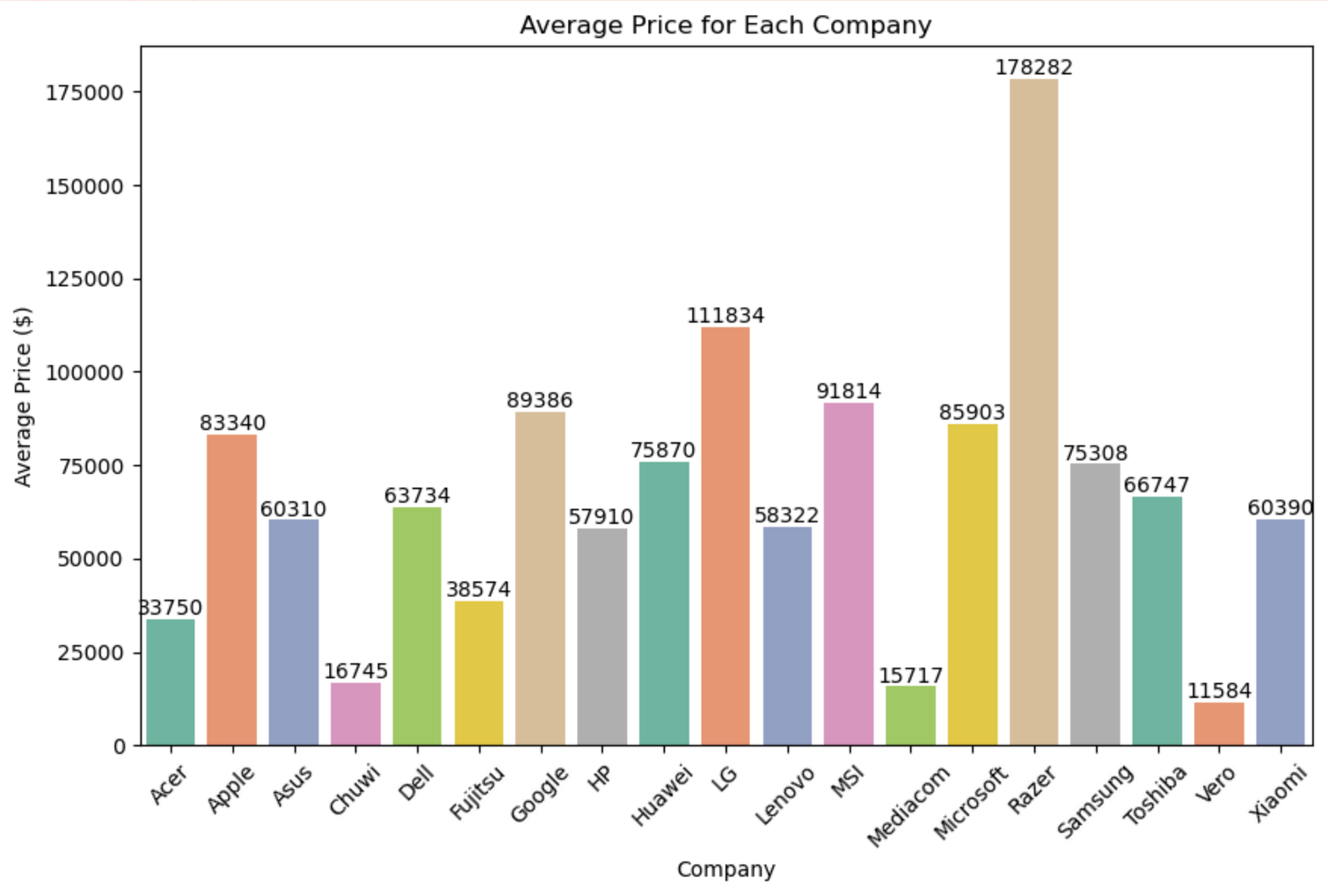


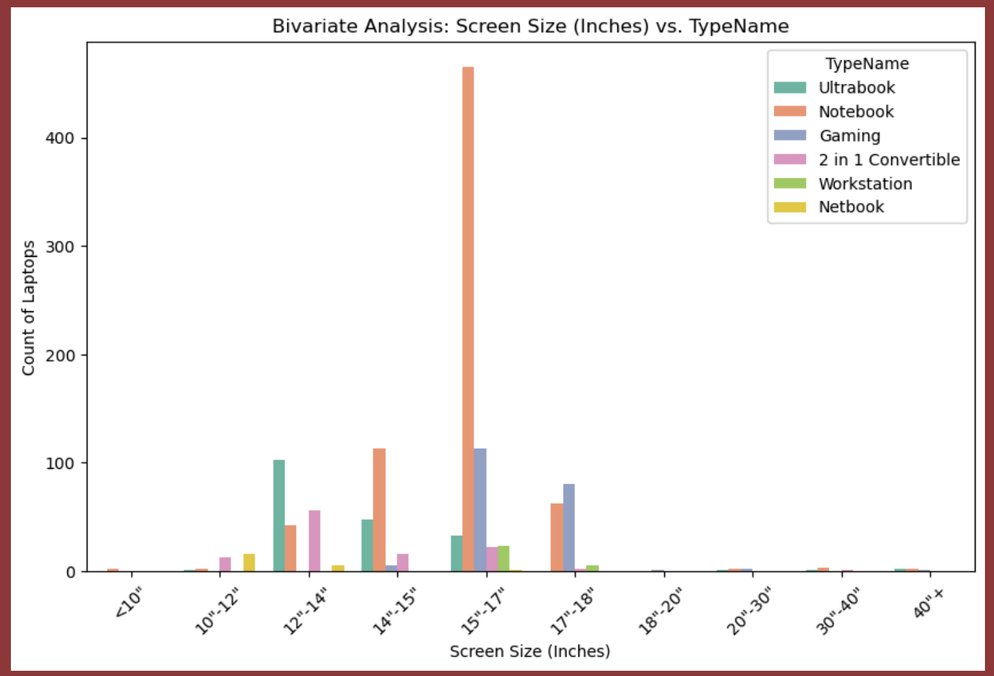
RAM amd its price range



weight and size

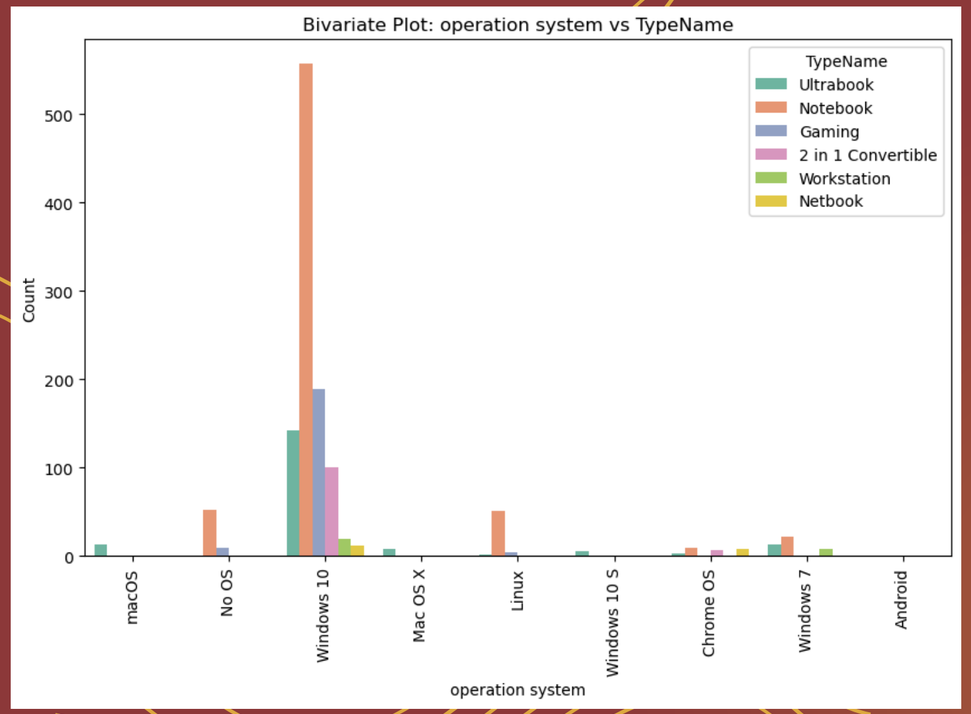

count
Memory
400
209
121
112
92
72
71
35
16
14
13
13
10
8
7
7
6
4
4
3
3
2
2
2
2
2
1
1
1


Average Price for Each Company
Average Price ($)
0
25000
50000
75000
100000
125000
150000
175000
33750
83340
60310
16745
63734
38574
89386
57910
75870
111834
58322
91814
15717
85903
178282
75308
66747
11584
60390
Acer
Apple
Asus
Chuwi
Dell
Fujitsu
Google
HP
Huawei
LG
Lenovo
MSI
Mediacom
Microsoft
Razer
Samsung
Toshiba
Vero
Xiaomi
Company

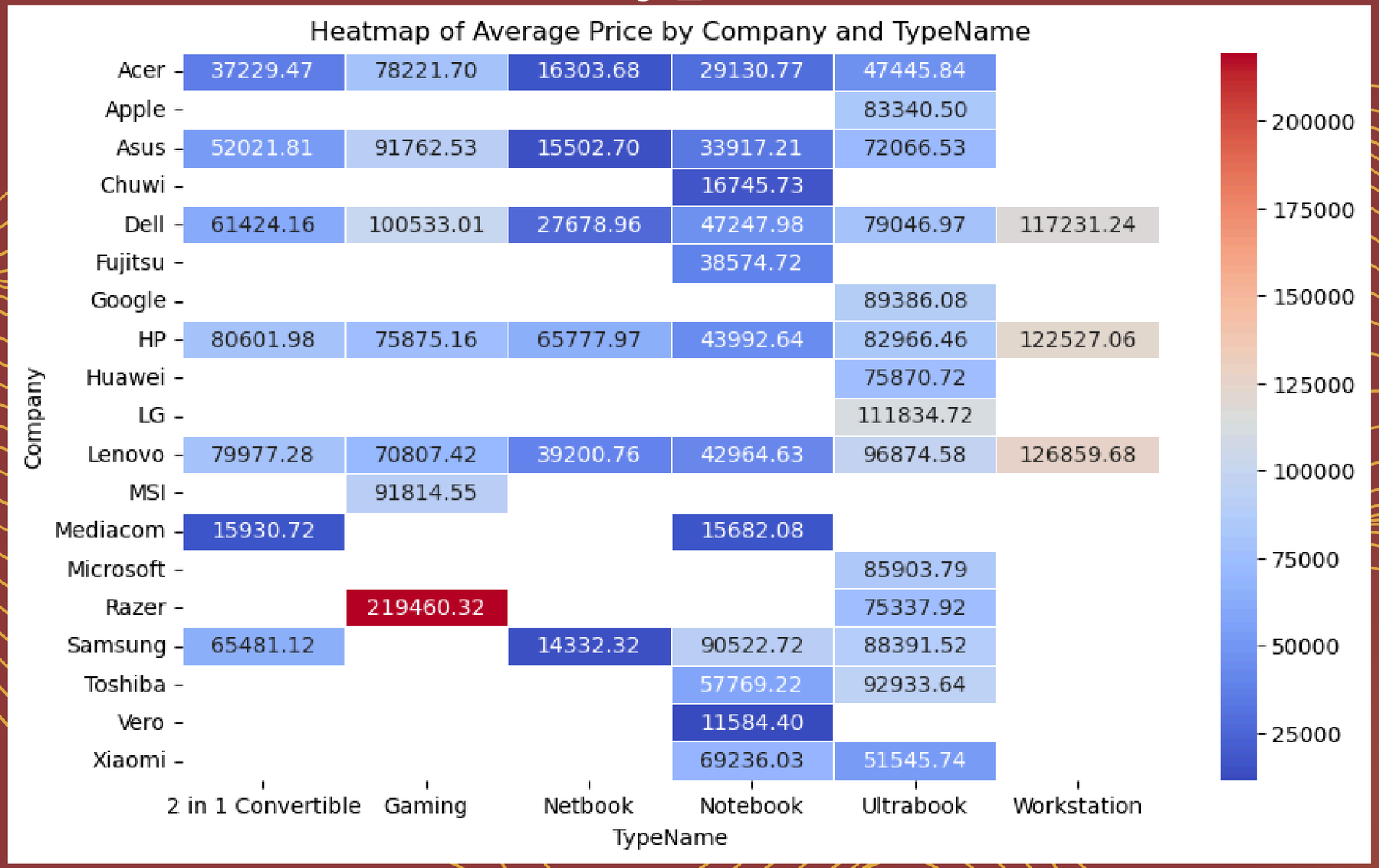


Count of typename of laptop on the basis of Inches

Count of typename of laptop on the basis of operation system



Price of laptop on the basis of Company and Type name



Thank you